



Viewing A29_s2m0FH- multi.table



Duke Biochemistry
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All-Atom Contacts	Clashscore, all atoms:	30.78		15 th percentile* (N=1784, all resolutions)
	Clashscore is the number of serious steric overlaps (> 0.4 Å) per 1000 atoms.			
Protein Geometry	Poor rotamers	0	0.00%	Goal: <0.3%
	Favored rotamers	287	99.65%	Goal: >98%
	Ramachandran outliers	0	0.00%	Goal: <0.05%
	Ramachandran favored	324	100.00%	Goal: >98%
	Rama distribution Z-score	5.09 ± 0.40		Goal: abs(Z score) < 2
	MolProbity score [^]	1.97		77 th percentile* (N=27675, 0Å - 99Å)
	Cβ deviations >0.25Å	0	0.00%	Goal: 0
	Bad bonds:	5 / 2676	0.19%	Goal: 0%
	Bad angles:	5 / 3597	0.14%	Goal: <0.1%
Peptide Omegas	Cis Prolines:	0 / 12	0.00%	Expected: ≤1 per chain, or ≤5%
Low-resolution Criteria	CaBLAM outliers	0	0.0%	Goal: <1.0%
	CA Geometry outliers	0	0.00%	Goal: <0.5%
Additional validations	Chiral volume outliers	0/408		
	Waters with clashes	0/0	0.00%	See UnDowser table for details

In the two column results, the left column gives the raw count, right column gives the percentage.

* 100th percentile is the best among structures of comparable resolution; 0th percentile is the worst. For clashscore the comparative set of structures was selected in 2004, for MolProbity score in 2006.

[^] MolProbity score combines the clashscore, rotamer, and Ramachandran evaluations into a single score, normalized to be on the same scale as X-ray resolution.

Key to table colors and cutoffs here: [?](#)

#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327
A 1		MET 34.25	0.54Å SD with A 4 THR O		-	Favored (43.7%) <i>ttm</i> chi angles: 187.7,174,276.7	0.01Å	-	-	-	-
A 2		ASP 41.96		-	Favored (25.74%) General / -91.8,11.1	Favored (46.6%) <i>p0</i> chi angles: 58.8,3.6	0.02Å	-	-	-	-
A 3		GLY 43.32		-	Favored (38.17%) Glycine / 106.3,3.9	-	-	Favored (51.949%)	-	-	-
A 4		THR 43.67	0.54Å O with A 1 MET SD		Favored (18.26%) General / -102.8,155.5	Favored (66.9%) <i>p</i> chi angles: 62.9	0.01Å	Favored (32.636%)	-	-	-
A 5		LEU 43.75		-	Favored (59.45%) General / -75.8,-31.7	Favored (58.9%) <i>mt</i> chi angles: 288.2,175.6	0.01Å	Favored (36.141%)	-	-	-
A 6		PHE 40.93		-	Favored (25.7%) Pre-Pro / -122.1,135.8	Favored (58.4%) <i>m-80</i> chi angles: 306.2,284.9	0.01Å	Favored (23.984%)	-	-	-

A 7	PRO	45.81	-		Favored (49.82%) Trans-Pro / -62.6,134.9	Favored (46.9%) <i>Cg_exo</i> chi angles: 338,33.6,328.7	0.04Å	Favored (28.247%)	-	-	-
A 8	GLY	44.9	-		Favored (85.06%) Glycine / 87.9,-1.3	-	-	Favored (81.383%)	-	-	-
A 9	ASP	48.28	-		Favored (92.76%) General / -59.4,-44.0	Favored (14.4%) <i>m-30</i> chi angles: 270,357.4	0.01Å	Favored (37.191%) alpha helix	-	-	-
A 10	ASP	50.42	-		Favored (83.69%) General / -66.5,-36.8	Favored (92.2%) <i>m-30</i> chi angles: 288.8,339.6	0.01Å	Favored (91.166%) alpha helix	-	-	-
A 11	ASP	50.3	-		Favored (69.04%) General / -69.2,-30.8	Favored (94.4%) <i>m-30</i> chi angles: 292.3,341	0.01Å	Favored (77.317%) alpha helix	-	-	-
A 12	LEU	48.86	-		Favored (44.67%) General / -98.3,8.6	Favored (86.8%) <i>mt</i> chi angles: 299.5,178.6	0.00Å	Favored (36.227%)	-	-	-
A 13	ALA	50.33	-		Favored (47.84%) General / -64.5,131.7	-	0.02Å	Favored (23.374%)	-	-	-
A 14	ILE	45.3	0.63Å CG2 with A 19 PHE HE1		Favored (17.86%) Pre-Pro / -130.3,117.3	Favored (62.4%) <i>mt</i> chi angles: 303.9,174.2	0.02Å	Favored (47.864%) beta sheet	-	-	-
A 15	PRO	49.85	0.70Å HB3 with C 110 GLU O		Favored (82%) Trans-Pro / -59.9,138.4	Favored (52.7%) <i>Cg_exo</i> chi angles: 336.9,34.3,328.7	0.03Å	Favored (69.3%)	-	-	-
A 16	ALA	44.36	-		Favored (70.98%) General / -60.5,-31.9	-	0.02Å	Favored (56.262%)	-	-	-
A 17	THR	48.88	-		Favored (67.7%) General / -63.5,-24.2	Favored (46.4%) <i>p</i> chi angles: 55.5	0.03Å	Favored (69.506%) three-ten	-	-	-
A 18	GLU	48.8	-		Favored (59.08%) General / -82.3,-10.4	Favored (90.1%) <i>mt-10</i> chi angles: 294.8,183.1,8.3	0.01Å	Favored (61.74%) three-ten	-	-	-
A 19	PHE	42.72	0.63Å HE1 with A 14 ILE CG2		Favored (65.22%) General / -68.7,-26.3	Favored (8.8%) <i>m-10</i> chi angles: 280.6,14	0.06Å	Favored (49.085%)	-	-	-
A 20	PHE	42.2	0.58Å HZ with C 93 MET HB3		Favored (2.14%) General / -132.4,51.6	Favored (58.9%) <i>m-80</i> chi angles: 306.3,281.3	0.02Å	Favored (17.977%)	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327
A 21	SER	43.6	-		Favored (37.07%) General / -75.7,153.5	Favored (51.4%) <i>p</i> chi angles: 56.5	0.02Å	Favored (15.254%)	-	-	-
A 22	THR	40.4	-		Favored (48.52%) General / -69.3,134.0	Favored (99%) <i>m</i> chi angles: 300.6	0.05Å	Favored (35.318%)	-	-	-

A 23	LYS	41.05	-	Favored (14.58%) General / -95.9,160.2	Favored (27.1%) <i>mm</i> chi angles: 296.6,293,179.7,283.6	0.02Å	Favored (22.8%)	-	-	-
A 24	ALA	37.93	-	Favored (57.6%) General / -66.8,138.8	-	0.02Å	Favored (35.186%)	-	-	-
A 25	ALA	45.87	-	Favored (14.23%) General / -95.7,161.0	-	0.02Å	Favored (27.84%)	-	-	-
A 26	LYS	48.31	0.82Å HB2 with A 31 LYS HE3	Favored (14.57%) General / -62.8,160.4	Favored (36.5%) <i>mtt</i> chi angles: 267.2,186.6,175.5,181.1	0.02Å	Favored (14.758%)	-	-	-
A 27	ASN	52.88	-	Favored (57.97%) Pre-Pro / -61.0,120.9	Favored (58.5%) <i>t</i> chi angles: 186.8,350.8	0.04Å	Favored (22.658%)	-	-	-
A 28	PRO	54.98	-	Favored (93.66%) Trans-Pro / -60.4,-34.2	Favored (54.3%) <i>Cg_exo</i> chi angles: 337.3,33.9,329	0.02Å	Favored (80.92%)	-	-	-
A 29	GLU	58.62	-	Favored (93%) General / -63.9,-39.0	Favored (66.7%) <i>tp30</i> chi angles: 184,64.9,22.1	0.01Å	Favored (77.598%) alpha helix	-	-	-
A 30	THR	63.68	-	Favored (99.38%) General / -61.5,-43.1	Favored (98.4%) <i>m</i> chi angles: 300.2	0.04Å	Favored (96.686%) alpha helix	-	-	-
A 31	LYS	59.25	0.82Å HE3 with A 26 LYS HB2	Favored (87.43%) General / -67.0,-40.0	Favored (99.5%) <i>mtt</i> chi angles: 294.2,178.6,179.1,176.4	0.01Å	Favored (95.568%) alpha helix	-	-	-
A 32	ARG	61.51	0.56Å HG2 with B 35 ILE HG12	Favored (99.18%) General / -63.3,-42.6	Favored (96.2%) <i>mtm-85</i> chi angles: 285.9,191.3,287.8,270.5	0.00Å	Favored (96.939%) alpha helix	-	-	-
A 33	GLU	64.7	-	Favored (89.94%) General / -65.8,-38.7	Favored (99.6%) <i>mt-10</i> chi angles: 292.5,178.1,352.8	0.01Å	Favored (95.121%) alpha helix	-	-	-
A 34	ALA	61.89	-	Favored (88.72%) General / -63.8,-37.8	-	0.02Å	Favored (90.745%) alpha helix	-	-	-
A 35	ILE	61.99	0.52Å CG2 with C 72 CYS SG	Favored (82.36%) Ile or Val / -68.6,-45.0	Favored (98.1%) <i>mt</i> chi angles: 292.6,168.7	0.05Å	Favored (84.683%) alpha helix	-	-	-
A 36	VAL	60.48	0.56Å HG11 with A 44 GLU N	Favored (58.04%) Ile or Val / -63.8,-32.7	Favored (68.7%) <i>t</i> chi angles: 172	0.06Å	Favored (76.132%) alpha helix	-	-	-
A 37	LYS	60.47	-	Favored (28.4%) General / -94.5,12.1	Favored (98.9%) <i>mtt</i> chi angles: 293.6,182.1,175.4,179.7	0.01Å	Favored (36.865%) alpha helix	-	-	-
A 38	ALA	55.72	-	Favored (90.15%) General / -61.9,-38.9	-	0.02Å	Favored (19.26%) alpha helix	-	-	-
A 39	TYR	52.76	-	Favored (26.26%) General / -128.6,120.2	Favored (86.9%) <i>m-80</i> chi angles: 301.3,279.9	0.03Å	CaBLAM Disfavored (1.823%)	-	-	-

A 40		GLY	56.59	-	Favored (41.74%) Glycine / 92.4,173.6	-	-	Favored (32.697%)	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non- Trans: 0 of 327
A 41		ASP	57.58	-	Favored (26.18%) General / -89.8,-18.5	Favored (69.8%) <i>m</i> - 30 chi angles: 298.9,332.6	0.03Å	CaBLAM Disfavored (3.605%)	-	-	-
A 42		ASP	57.93	0.86Å OD2 with A 50 ARG NH1	Favored (24.08%) General / -86.8,150.5	Favored (78.6%) <i>m</i> - 30 chi angles: 295,328	0.02Å	Favored (18.827%)	-	-	-
A 43		ASN	61.35	0.41Å HA with A 36 VAL HG12	Favored (3.43%) General / -101.9,-177.5	Favored (44.6%) <i>p</i> 0 chi angles: 60.2,13	0.04Å	Favored (26.214%)	-	-	-
A 44		GLU	63.74	0.56Å N with A 36 VAL HG11	Favored (95.55%) General / -62.5,-44.8	Favored (31%) <i>tt</i> 0 chi angles: 179.1,191.8,320	0.04Å	Favored (45.641%)	-	-	-
A 45		GLU	68.34	-	Favored (93.62%) General / -64.5,-39.3	Favored (70.5%) <i>tp</i> 30 chi angles: 179.2,65.6,14.7	0.02Å	Favored (92.411%) alpha helix	-	-	-
A 46		THR	71.81	-	Favored (94.5%) General / -64.9,-40.0	Favored (90.9%) <i>m</i> chi angles: 298.9	0.04Å	Favored (96.063%) alpha helix	-	-	-
A 47		LEU	72.48	-	Favored (99.18%) General / -63.2,-41.2	Favored (72.6%) <i>mt</i> chi angles: 287.5,172	0.01Å	Favored (98.061%) alpha helix	-	-	-
A 48		LYS	73.71	-	Favored (99.31%) General / -63.2,-42.1	Favored (73.1%) <i>mttt</i> chi angles: 292.9,183.5,178.5,197.2	0.01Å	Favored (98.867%) alpha helix	-	-	-
A 49		GLN	77.83	-	Favored (94.01%) General / -65.1,-40.1	Favored (96%) <i>mt</i> 0 chi angles: 293.5,172.4,344.2	0.01Å	Favored (97.932%) alpha helix	-	-	-
A 50		ARG	78.54	0.86Å NH1 with A 42 ASP OD2	Favored (94.33%) General / -65.3,-41.2	Favored (88.5%) <i>mtp</i> 180 chi angles: 290.8,175.1,65.2,188.6	0.01Å	Favored (99.457%) alpha helix	-	-	-
A 51		LEU	76.96	0.61Å HD21 with B 51 LEU CD2	Favored (99.03%) General / -63.3,-42.6	Favored (62.9%) <i>mt</i> chi angles: 285.9,168.7	0.01Å	Favored (99.408%) alpha helix	-	-	-
A 52		THR	78.92	-	Favored (99.8%) General / -63.0,-42.4	Favored (88.5%) <i>m</i> chi angles: 298.4	0.04Å	Favored (99.091%) alpha helix	-	-	-
A 53		ASN	80.92	-	Favored (96.2%) General / -64.3,-40.6	Favored (98.9%) <i>m</i> - 40 chi angles: 289.3,340.3	0.02Å	Favored (98.811%) alpha helix	-	-	-
A 54		LEU	78.24	-	Favored (88.43%) General / -65.8,-38.3	Favored (5.9%) <i>mp</i> chi angles: 265.5,60.8	0.02Å	Favored (92.844%) alpha helix	-	-	-

A 55	GLU 80.37			1.04Å HG2 with B 54 LEU HD11	Favored (96.59%) General / -60.5,-42.5	Favored (96.5%) <i>mt-10</i> chi angles: 290.5,179.5,4.4	0.01Å	Favored (93.327%) alpha helix	-	-	-
A 56		LYS	82.79	-	Favored (95.48%) General / -64.9,-41.5	Favored (98.9%) <i>mttt</i> chi angles: 293.2,178.7,181.1,176.7	0.01Å	Favored (97.421%) alpha helix	-	-	-
A 57	LYS 78.76			-	Favored (96.01%) General / -64.5,-42.3	Favored (62.9%) <i>mttm</i> chi angles: 292.8,179.7,183.5,293.5	0.01Å	Favored (98.995%) alpha helix	-	-	-
A 58		ILE	77.17	0.43Å HD11 with C 58 ILE HD11	Favored (84.89%) Ile or Val / -65.6,-39.4	Favored (81.6%) <i>mt</i> chi angles: 291,172.6	0.07Å	Favored (97.402%) alpha helix	-	-	-
A 59	THR 80.63			-	Favored (99.46%) General / -62.1,-43.1	Favored (91.2%) <i>m</i> chi angles: 298.9	0.05Å	Favored (97.258%) alpha helix	-	-	-
A 60		ASN	78.54	-	Favored (95.33%) General / -64.9,-40.6	Favored (98.6%) <i>m-40</i> chi angles: 291.3,337.8	0.01Å	Favored (95.969%) alpha helix	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non- Trans: 0 of 327
A 61	ILE 75.01			-	Favored (85.97%) Ile or Val / -65.9,-39.9	Favored (96.2%) <i>mt</i> chi angles: 293.2,169.6	0.06Å	Favored (92.671%) alpha helix	-	-	-
A 62		THR	76.28	-	Favored (97.12%) General / -60.9,-42.0	Favored (98%) <i>m</i> chi angles: 300.1	0.04Å	Favored (90.459%) alpha helix	-	-	-
A 63	THR 77.82			-	Favored (97.8%) General / -61.5,-44.0	Favored (96.2%) <i>m</i> chi angles: 299.7	0.05Å	Favored (97.971%) alpha helix	-	-	-
A 64		LYS	73.24	-	Favored (97.17%) General / -63.5,-40.5	Favored (93.4%) <i>mttt</i> chi angles: 288.8,186.6,176,183.1	0.01Å	Favored (96.006%) alpha helix	-	-	-
A 65	PHE 74.16			0.74Å CE1 with B 65 PHE HD1	Favored (97.72%) General / -63.1,-40.5	Favored (4.8%) <i>m-80</i> chi angles: 266.7,291.8	0.00Å	Favored (90.666%) alpha helix	-	-	-
A 66		GLU	76.45	-	Favored (90.54%) General / -64.5,-38.3	Favored (99.8%) <i>mt-10</i> chi angles: 292.1,175.2,350.4	0.02Å	Favored (92.341%) alpha helix	-	-	-
A 67	GLN 74.46			-	Favored (83.34%) General / -67.9,-40.1	Favored (98.5%) <i>mt0</i> chi angles: 292.8,176.4,325.6	0.01Å	Favored (96.201%) alpha helix	-	-	-
A 68		ILE	72.17	-	Favored (86.67%) Ile or Val / -66.0,-40.2	Favored (96%) <i>mt</i> chi angles: 293,169.9	0.06Å	Favored (96.029%) alpha helix	-	-	-
A 69	GLU 74.04			-	Favored (98.94%) General / -62.4,-41.8	Favored (83%) <i>tt0</i> chi angles: 184,181,11.9	0.01Å	Favored (95.405%) alpha helix	-	-	-

A 70	LYS	75.74	-		Favored (97.45%) General / -61.0,-41.9	Favored (83.6%) <i>tttt</i> chi angles: 178.1,180.3,177.3,181.5	0.03 Å	Favored (98.879%) alpha helix	-	-	-
A 71	CYS	72.88	0.68 Å HB2 with C 39 TYR HE2		Favored (97.78%) General / -62.6,-43.8	Favored (36.3%) <i>t</i> chi angles: 174.6	0.03 Å	Favored (98.238%) alpha helix	-	-	-
A 72	CYS	74.31	1.21 Å HB3 with B 71 CYS SG		Favored (98.57%) General / -63.5,-41.5	Favored (98.6%) <i>m</i> chi angles: 289.7	0.01 Å	Favored (94.709%) alpha helix	-	-	-
A 73	LYS	74.08	-		Favored (87.2%) General / -66.7,-38.7	Favored (97.4%) <i>mttt</i> chi angles: 289.9,182.1,176.9,178.2	0.02 Å	Favored (96.741%) alpha helix	-	-	-
A 74	HIS	73.73	-		Favored (92.43%) General / -65.8,-40.8	Favored (88.1%) <i>t70</i> chi angles: 182.6,73.2	0.01 Å	Favored (95.697%) alpha helix	-	-	-
A 75	ASN	73.43	-		Favored (87.45%) General / -61.9,-38.2	Favored (88.3%) <i>m-40</i> chi angles: 286.8,349.2	0.01 Å	Favored (86.153%) alpha helix	-	-	-
A 76	ASP	72.37	-		Favored (91.47%) General / -58.9,-44.9	Favored (12.1%) <i>m-30</i> chi angles: 273,333.9	0.03 Å	Favored (88.552%) alpha helix	-	-	-
A 77	GLU	74.16	-		Favored (95.85%) General / -63.2,-44.3	Favored (85.5%) <i>tt0</i> chi angles: 178.2,181.8,358.5	0.03 Å	Favored (89.526%) alpha helix	-	-	-
A 78	VAL	71.71	-		Favored (90.14%) Ile or Val / -66.7,-43.5	Favored (93.2%) <i>t</i> chi angles: 174.6	0.05 Å	Favored (90.467%) alpha helix	-	-	-
A 79	LEU	71.64	-		Favored (88.68%) General / -64.5,-37.8	Favored (89.1%) <i>mt</i> chi angles: 290.7,171.9	0.02 Å	Favored (88.744%) alpha helix	-	-	-
A 80	PHE	73.04	-		Favored (98.18%) General / -61.4,-43.8	Favored (87.4%) <i>t80</i> chi angles: 180.5,80.5	0.03 Å	Favored (91.864%) alpha helix	-	-	-
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A 81	ARG	73.37	-		Favored (92.42%) General / -64.3,-44.2	Favored (16.8%) <i>ttm110</i> chi angles: 188.9,185.6,317.3,129.3	0.04 Å	Favored (93.25%) alpha helix	-	-	-
A 82	LEU	71.62	-		Favored (85.5%) General / -65.3,-36.9	Favored (82.1%) <i>mt</i> chi angles: 290.5,174.2	0.01 Å	Favored (88.277%) alpha helix	-	-	-
A 83	GLU	71.49	1.06 Å HA with B 82 LEU HD21		Favored (98.08%) General / -63.7,-41.2	Favored (91.1%) <i>tt0</i> chi angles: 181.3,180,6.7	0.01 Å	Favored (92.818%) alpha helix	-	-	-
A 84	ASN	73.8	-		Favored (96.13%) General / -64.4,-42.3	Favored (98.7%) <i>m-40</i> chi angles: 289.8,341.8	0.01 Å	Favored (97.312%) alpha helix	-	-	-

A 85	HIS	71.75	-		Favored (95.19%) General / -64.9,-40.5	Favored (15.9%) <i>m-70</i> chi angles: 287.8,322	0.01Å	Favored (97.251%) alpha helix	-	-	-
A 86	ALA	70.59	-		Favored (98.23%) General / -63.6,-42.5	-	0.02Å	Favored (96.937%) alpha helix	-	-	-
A 87	GLU	73.16	-		Favored (90.84%) General / -65.6,-38.9	Favored (67.8%) <i>tp30</i> chi angles: 183.8,65.6,18.3	0.02Å	Favored (96.689%) alpha helix	-	-	-
A 88	THR	73.33	-		Favored (92.55%) General / -65.8,-40.9	Favored (38.6%) <i>m</i> chi angles: 305.3	0.05Å	Favored (99.073%) alpha helix	-	-	-
A 89	LEU	72.68	-		Favored (82.63%) General / -65.5,-36.0	Favored (70.2%) <i>mt</i> chi angles: 287,170.4	0.01Å	Favored (87.748%) alpha helix	-	-	-
A 90	ARG	71.32	1.24Å HG3 with C 20 PHE CE1		Favored (89.88%) General / -61.7,-46.4	Favored (79%) <i>ttp80</i> chi angles: 184.4,191.6,64.3,83.3	0.02Å	Favored (83.82%) alpha helix	-	-	-
A 91	ALA	74.85	-		Favored (94.29%) General / -64.3,-39.6	-	0.02Å	Favored (89.403%) alpha helix	-	-	-
A 92	ALA	71.15	-		Favored (91%) General / -64.3,-38.4	-	0.02Å	Favored (90.772%) alpha helix	-	-	-
A 93	MET	70.66	-		Favored (95.65%) General / -63.9,-40.1	Favored (71.5%) <i>mmm</i> chi angles: 285,288.8,292.4	0.01Å	Favored (96.662%) alpha helix	-	-	-
A 94	ILE	72.32	-		Favored (96.58%) Ile or Val / -63.4,-46.0	Favored (92.8%) <i>mt</i> chi angles: 293.6,172.4	0.06Å	Favored (93.807%) alpha helix	-	-	-
A 95	SER	74.05	-		Favored (94.23%) General / -64.1,-39.5	Favored (69.5%) <i>m</i> chi angles: 296.5	0.01Å	Favored (92.813%) alpha helix	-	-	-
A 96	LEU	71.75	-		Favored (90.09%) General / -66.2,-39.8	Favored (53.6%) <i>tp</i> chi angles: 182.8,59.3	0.01Å	Favored (93.619%) alpha helix	-	-	-
A 97	ALA	70.41	-		Favored (98.39%) General / -61.3,-42.3	-	0.02Å	Favored (95.89%) alpha helix	-	-	-
A 98	LYS	74.02	0.46Å NZ with A 109 TYR O		Favored (96.44%) General / -64.6,-41.2	Favored (62.6%) <i>mttm</i> chi angles: 292.8,180.8,179.2,294.8	0.02Å	Favored (95.985%) alpha helix	-	-	-
A 99	LYS	71.62	-		Favored (98.43%) General / -63.3,-40.8	Favored (83%) <i>tttt</i> chi angles: 178.3,181.5,177.7,182.3	0.02Å	Favored (95.892%) alpha helix	-	-	-
A 100	ILE	70.6	-		Favored (95.88%) Ile or Val / -63.8,-42.4	Favored (80.1%) <i>mt</i> chi angles: 290,169.5	0.06Å	Favored (97.819%) alpha helix	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327

A 101	ASP	69.42	0.56Å CG with B 99 LYS HZ1	Favored (99.19%) General / -61.4,-43.2	Favored (29.9%) <i>m-30</i> chi angles: 273.6,356.1	0.01Å	Favored (96.894%) alpha helix	-	-	-
A 102	VAL	71.72	-	Favored (95.02%) Ile or Val / -65.3,-44.4	Favored (58.9%) <i>t</i> chi angles: 170.6	0.06Å	Favored (91.813%) alpha helix	-	-	-
A 103	GLN	70.89	-	Favored (62.95%) General / -70.2,-23.7	Favored (16.4%) <i>mt0</i> chi angles: 290,182.3,168.3	0.04Å	Favored (70.674%) alpha helix	-	-	-
A 104	THR	67.12	0.99Å O with C 5 LEU HD12	Favored (41.02%) General / -99.7,10.6	Favored (66.1%) <i>p</i> chi angles: 58.4	0.03Å	Favored (48.919%)	-	-	-
A 105	GLY	68.2	-	Favored (58.68%) Glycine / 85.9,19.0	-	-	Favored (82.362%)	-	-	-
A 106	ARG	62.22	0.44Å HD2 with C 9 ASP OD2	Favored (22.63%) General / -113.6,157.0	Favored (72.6%) <i>mtp180</i> chi angles: 297.9,189.2,61.2,186.5	0.02Å	Favored (23.966%)	-	-	-
A 107	ARG	57.44	-	Favored (86.7%) Pre-Pro / -74.5,149.9	Favored (42.5%) <i>ptt-90</i> chi angles: 72.4,184.2,192.3,276.6	0.00Å	Favored (40.035%)	-	-	-
A 108	PRO	55.19	-	Favored (51.38%) Trans-Pro / -64.6,158.2	Favored (36.1%) <i>Cg_exo</i> chi angles: 338.9,33.5,327.8	0.04Å	Favored (66.329%)	-	-	-
A 109	TYR	56.37	0.64Å CE2 with B 99 LYS NZ	Favored (45.57%) General / -70.0,151.4	Favored (34.6%) <i>m-80</i> chi angles: 287.1,301.5	0.01Å	-	-	-	-
A 110	GLU	51.41	-	-	Favored (5.2%) <i>tt0</i> chi angles: 207.6,172.7,35.6	0.02Å	-	-	-	-
B 1	MET	33.79	0.41Å SD with B 4 THR O	-	Favored (37.8%) <i>ttm</i> chi angles: 188.3,174.8,274	0.02Å	-	-	-	-
B 2	ASP	43.03	-	Favored (25.4%) General / -92.2,11.5	Favored (43.4%) <i>p0</i> chi angles: 57.6,3.9	0.01Å	-	-	-	-
B 3	GLY	42.36	-	Favored (42.58%) Glycine / 104.6,-1.3	-	-	Favored (49.63%)	-	-	-
B 4	THR	42.83	0.41Å O with B 1 MET SD	Favored (19.34%) General / -106.6,155.7	Favored (69.2%) <i>p</i> chi angles: 62.5	0.02Å	Favored (33.529%)	-	-	-
B 5	LEU	43.54	-	Favored (58.58%) General / -76.2,-31.4	Favored (65.3%) <i>mt</i> chi angles: 288.7,175.1	0.01Å	Favored (34.163%)	-	-	-
B 6	PHE	39.91	0.57Å HB3 with C 103 GLN HA	Favored (25.87%) Pre-Pro / -124.0,135.0	Favored (54.9%) <i>m-80</i> chi angles: 307,284.2	0.01Å	Favored (24.418%)	-	-	-
B 7	PRO	46.46	0.58Å O with C 102 VAL HG12	Favored (58.08%) Trans-Pro / -61.8,135.8	Favored (43.2%) <i>Cg_exo</i> chi angles: 338.2,33.6,328.5	0.03Å	Favored (27.624%)	-	-	-
B 8	GLY	45.68	-	Favored (85.33%) Glycine / 87.7,-1.0	-	-	Favored (79.795%)	-	-	-

B 9	ASP 47.04			0.67Å OD1 with C 103 GLN HG2	Favored (93.09%) General / -59.4,-43.3	Favored (17.9%) <i>m</i> - 30 chi angles: 271.6,1.1	0.01Å	Favored (37.667%) alpha helix	-	-	-
B 10	ASP 48.99			-	Favored (87.9%) General / -65.1,-37.5	Favored (93.7%) <i>m</i> - 30 chi angles: 289.4,340.4	0.01Å	Favored (95.444%) alpha helix	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non- Trans: 0 of 327
B 11	ASP 49.39			-	Favored (70.4%) General / -69.3,-31.7	Favored (95%) <i>m</i> - 30 chi angles: 292.2,341.6	0.01Å	Favored (75.725%) alpha helix	-	-	-
B 12	LEU 47.63			0.72Å HD13 with C 99 LYS HA	Favored (44.83%) General / -99.7,7.8	Favored (86.3%) <i>mt</i> chi angles: 299.5,178.8	0.01Å	Favored (36.035%)	-	-	-
B 13	ALA 49.48			-	Favored (51.48%) General / -64.8,133.2	-	0.02Å	Favored (24.902%)	-	-	-
B 14	ILE 41.92			1.04Å HD11 with C 95 SER HB3	Favored (18.59%) Pre-Pro / -129.2,115.2	Favored (61.6%) <i>mt</i> chi angles: 304,174.6	0.02Å	Favored (52.312%) beta sheet	-	-	-
B 15	PRO 46.13			-	Favored (82.12%) Trans-Pro / -62.5,140.4	Favored (52.7%) <i>Cg_exo</i> chi angles: 337.6,33.8,328.7	0.03Å	Favored (60.92%)	-	-	-
B 16	ALA 40.25			-	Favored (70.11%) General / -59.3,-32.5	-	0.01Å	Favored (57.935%)	-	-	-
B 17	THR 45			-	Favored (68.12%) General / -63.8,-25.3	Favored (45%) <i>p</i> chi angles: 55.3	0.03Å	Favored (70.049%) three-ten	-	-	-
B 18	GLU 44.85			-	Favored (59.75%) General / -80.1,-10.1	Favored (91.1%) <i>mt-10</i> chi angles: 294.6,182.7,8	0.01Å	Favored (60.274%) three-ten	-	-	-
B 19	PHE 38.69			0.64Å HE1 with B 14 ILE CG2	Favored (63.32%) General / -70.0,-18.7	Favored (7.9%) <i>m</i> - 10 chi angles: 279.1,14.9	0.08Å	Favored (45.543%)	-	-	-
B 20	PHE 37.25			0.52Å HA with C 88 THR HG21	Favored (2.04%) General / -138.9,44.4	Favored (53.6%) <i>m</i> - 80 chi angles: 307.2,281.4	0.02Å	Favored (17.315%)	-	-	-
B 21	SER 40.62			-	Favored (33.24%) General / -78.3,150.7	Favored (50.8%) <i>p</i> chi angles: 56.5	0.02Å	Favored (22.341%)	-	-	-
B 22	THR 38.75			-	Favored (43.33%) General / -73.8,134.4	Favored (98.4%) <i>m</i> chi angles: 300.1	0.05Å	Favored (38.405%)	-	-	-
B 23	LYS 40.27			-	Favored (14.1%) General / -100.3,160.7	Favored (27.3%) <i>mmtm</i> chi angles: 298,291.8,179.9,284.2	0.02Å	Favored (22.766%)	-	-	-
B 24	ALA 40.16			-	Favored (55.54%) General / -67.0,136.6	-	0.02Å	Favored (32.214%)	-	-	-

B 25	ALA	44.58	-		Favored (17.36%) General / -103.7,157.0	-	0.02Å	Favored (33.269%)	-	-	-
B 26	LYS	48.24	0.41Å HB2 with B 31 LYS HE3		Favored (29.91%) General / -67.3,159.1	Favored (51.4%) <i>mttt</i> chi angles: 270.4,185.6,175.2,180.4	0.03Å	Favored (26.977%)	-	-	-
B 27	ASN	53.72	-		Favored (59.42%) Pre-Pro / -64.6,119.7	Favored (58.2%) <i>t0</i> chi angles: 186.5,351.8	0.05Å	Favored (23.334%)	-	-	-
B 28	PRO	54.1	0.61Å HG3 with C 80 PHE HE2		Favored (81.91%) Trans-Pro / -61.9,-34.3	Favored (36.7%) <i>Cg_exo</i> chi angles: 338.8,32.9,329.2	0.03Å	Favored (69.989%)	-	-	-
B 29	GLU	58.19	-		Favored (87.04%) General / -63.9,-37.3	Favored (68.7%) <i>tp30</i> chi angles: 183.9,66.5,19.7	0.01Å	Favored (75.631%) alpha helix	-	-	-
B 30	THR	61.51	-		Favored (98.92%) General / -61.4,-43.4	Favored (95.8%) <i>m</i> chi angles: 299.6	0.04Å	Favored (88.158%) alpha helix	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non- Trans: 0 of 327
B 31	LYS	55.42	0.75Å NZ with C 80 PHE HE1		Favored (81.18%) General / -68.4,-40.3	Favored (99.6%) <i>mttt</i> chi angles: 294.2,179.1,179.1,175.2	0.01Å	Favored (94.28%) alpha helix	-	-	-
B 32	ARG	58.87	-		Favored (96.92%) General / -64.2,-41.5	Favored (97.2%) <i>mtm-85</i> chi angles: 286,191,289.1,271.7	0.01Å	Favored (98.48%) alpha helix	-	-	-
B 33	GLU	62.44	-		Favored (85.35%) General / -66.2,-37.3	Favored (98.9%) <i>mt-10</i> chi angles: 290.8,179,355.3	0.01Å	Favored (88.211%) alpha helix	-	-	-
B 34	ALA	57.29	-		Favored (92.03%) General / -62.5,-39.0	-	0.02Å	Favored (84.175%) alpha helix	-	-	-
B 35	ILE	56.79	0.56Å HG12 with A 32 ARG HG2		Favored (84.69%) Ile or Val / -68.2,-43.8	Favored (96.9%) <i>mt</i> chi angles: 293.7,168.4	0.05Å	Favored (79.961%) alpha helix	-	-	-
B 36	VAL	54.99	0.53Å HG21 with B 44 GLU HB2		Favored (15.62%) Ile or Val / -75.5,-15.6	Favored (88.3%) <i>t</i> chi angles: 176.1	0.03Å	Favored (58.103%) alpha helix	-	-	-
B 37	LYS	56.33	0.58Å HE3 with B 43 ASN HB3		Favored (23.78%) General / -102.5,17.4	Favored (96.8%) <i>mttt</i> chi angles: 289.2,183.2,175.7,180.3	0.02Å	Favored (37.903%)	-	-	-
B 38	ALA	47.56	-		Favored (70.93%) General / -57.3,-36.3	-	0.02Å	Favored (15.833%)	-	-	-
B 39	TYR	47.2	-		Favored (42.08%) General / -134.6,135.3	Favored (93.8%) <i>m-80</i> chi angles: 292,87.2	0.06Å	CaBLAM Disfavored (2.02%)	OUTLIER(S) worst is C--O: 4.6 σ	-	-

B 40	GLY	53.76	-		Favored (54.58%) Glycine / 81.2,-174.2	-	-	Favored (33.513%)	-	-	-
B 41	ASP	57.86	-		Favored (23.93%) General / -94.1,-16.3	Favored (76.4%) <i>m-30</i> chi angles: 297.6,333.3	0.03Å	CaBLAM Disfavored (2.197%)	-	-	-
B 42	ASP	55.84	-		Favored (32.45%) General / -79.3,147.5	Favored (81.9%) <i>m-30</i> chi angles: 293.9,331.2	0.01Å	Favored (21.712%)	-	-	-
B 43	ASN	61.41	0.58Å HB3 with B 37 LYS HE3		Favored (2.9%) General / -109.1,-175.9	Favored (42.2%) <i>p0</i> chi angles: 59.7,12.2	0.04Å	Favored (21.677%)	-	-	-
B 44	GLU	62.54	0.53Å HB2 with B 36 VAL HG21		Favored (97.87%) General / -61.7,-43.9	Favored (64.5%) <i>tt0</i> chi angles: 178.5,183.9,337.6	0.03Å	Favored (43.196%)	-	-	-
B 45	GLU	69.76	-		Favored (91.44%) General / -64.7,-38.6	Favored (71.7%) <i>tp30</i> chi angles: 179.6,66.4,15.5	0.01Å	Favored (91.6%) alpha helix	-	-	-
B 46	THR	73.09	-		Favored (92.88%) General / -65.4,-40.0	Favored (88.8%) <i>m</i> chi angles: 298.6	0.04Å	Favored (96.968%) alpha helix	-	-	-
B 47	LEU	71.26	-		Favored (97.16%) General / -64.1,-41.6	Favored (74.1%) <i>mt</i> chi angles: 288.1,172.8	0.02Å	Favored (98.953%) alpha helix	-	-	-
B 48	LYS	72.52	0.42Å NZ with C 69 GLU OE2		Favored (98.68%) General / -60.9,-43.3	Favored (97.3%) <i>mttt</i> chi angles: 290.6,181.7,178.7,179.7	0.03Å	Favored (97.925%) alpha helix	-	-	-
B 49	GLN	79.13	-		Favored (95.32%) General / -64.1,-40.0	Favored (59.8%) <i>tt0</i> chi angles: 186.2,172.4,341.2	0.03Å	Favored (95.81%) alpha helix	-	-	-
B 50	ARG	78.58	-		Favored (95.78%) General / -64.7,-41.9	Favored (88%) <i>mtp180</i> chi angles: 290.8,175.7,64.8,189.8	0.01Å	Favored (98.88%) alpha helix	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327
B 51	LEU	75.57	0.61Å CD2 with A 51 LEU HD21		Favored (97.94%) General / -63.7,-42.6	Favored (62.5%) <i>mt</i> chi angles: 285.8,168.3	0.01Å	Favored (98.262%) alpha helix	-	-	-
B 52	THR	80.25	-		Favored (99.69%) General / -62.9,-42.1	Favored (88.5%) <i>m</i> chi angles: 298.5	0.04Å	Favored (99.162%) alpha helix	-	-	-
B 53	ASN	81.48	-		Favored (96.26%) General / -63.8,-40.3	Favored (99.1%) <i>m-40</i> chi angles: 288.8,341	0.02Å	Favored (98.104%) alpha helix	-	-	-
B 54	LEU	75.36	1.04Å HD11 with A 55 GLU HG2		Favored (90.07%) General / -65.6,-38.6	Favored (5.7%) <i>mp</i> chi angles: 265.1,61.5	0.02Å	Favored (96.871%) alpha helix	-	-	-
B 55	GLU	78.85	-		Favored (98.58%) General / -62.0,-41.9	Favored (99.1%) <i>mt-10</i> chi angles: 292,179.1,1.6	0.01Å	Favored (98.223%) alpha helix	-	-	-

B 56	LYS	83.11	-		Favored (95.05%) General / -64.5,-40.1	Favored (98.8%) <i>mttt</i> chi angles: 292.8,178.3,181.5,176.7	0.01Å	Favored (98.154%) alpha helix	-	-	-
B 57	LYS	78.87	-		Favored (97.4%) General / -63.9,-42.2	Favored (62.9%) <i>mttm</i> chi angles: 291.7,179.7,184.4,293.7	0.01Å	Favored (98.908%) alpha helix	-	-	-
B 58	ILE	74.4	0.81Å HG21 with C 55 GLU HG2		Favored (94.15%) Ile or Val / -62.2,-42.2	Favored (12.4%) <i>mm</i> chi angles: 287.5,290.5	0.05Å	Favored (99.219%) alpha helix	-	-	-
B 59	THR	81.45	-		Favored (98.78%) General / -62.3,-43.5	Favored (93.1%) <i>m</i> chi angles: 299.2	0.05Å	Favored (97.508%) alpha helix	-	-	-
B 60	ASN	79.55	-		Favored (94.08%) General / -64.9,-39.8	Favored (98.5%) <i>m-40</i> chi angles: 290.9,338.7	0.02Å	Favored (96.867%) alpha helix	-	-	-
B 61	ILE	73.15	0.70Å HG21 with C 51 LEU HD21		Favored (87.03%) Ile or Val / -65.6,-40.1	Favored (94.8%) <i>mt</i> chi angles: 292.1,169.4	0.06Å	Favored (97.166%) alpha helix	-	-	-
B 62	THR	76.44	-		Favored (98.56%) General / -61.6,-42.2	Favored (96.1%) <i>m</i> chi angles: 300.8	0.04Å	Favored (97.714%) alpha helix	-	-	-
B 63	THR	79.94	-		Favored (97.02%) General / -62.3,-44.2	Favored (96.1%) <i>m</i> chi angles: 299.7	0.04Å	Favored (97.97%) alpha helix	-	-	-
B 64	LYS	73.47	-		Favored (97.37%) General / -63.1,-40.4	Favored (89.2%) <i>mttt</i> chi angles: 286.6,188.4,175.6,183.8	0.01Å	Favored (93.006%) alpha helix	-	-	-
B 65	PHE	71.86	0.74Å HD1 with A 65 PHE CE1		Favored (98.81%) General / -62.6,-41.2	Favored (81.1%) <i>t80</i> chi angles: 182.6,84.2	0.01Å	Favored (90.422%) alpha helix	-	-	-
B 66	GLU	76.25	-		Favored (91.86%) General / -62.8,-38.8	Favored (99%) <i>mt-10</i> chi angles: 293,176,349.6	0.02Å	Favored (94.595%) alpha helix	-	-	-
B 67	GLN	75.58	-		Favored (85.2%) General / -67.5,-39.8	Favored (97.8%) <i>mt0</i> chi angles: 293.4,175.4,326.5	0.01Å	Favored (98%) alpha helix	-	-	-
B 68	ILE	70.16	0.55Å HD12 with A 65 PHE HD1		Favored (85.77%) Ile or Val / -65.1,-39.5	Favored (92.1%) <i>mt</i> chi angles: 291.8,169.7	0.06Å	Favored (95.259%) alpha helix	-	-	-
B 69	GLU	72.57	-		Favored (99.07%) General / -62.3,-42.0	Favored (85.2%) <i>tt0</i> chi angles: 185.3,179.9,11.5	0.01Å	Favored (94.134%) alpha helix	-	-	-
B 70	LYS	76.51	-		Favored (95.73%) General / -62.5,-40.1	Favored (97.3%) <i>mttt</i> chi angles: 290.1,180.8,177.6,179.3	0.01Å	Favored (97.954%) alpha helix	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327

B 71	CYS	72.05	1.21Å SG with A 72 CYS HB3	Favored (96.9%) General / -62.0,-44.4	Favored (34.5%) <i>t</i> chi angles: 174.4	0.03Å	Favored (97.418%) alpha helix	-	-	-
B 72	CYS	71.03	-	Favored (99.53%) General / -62.9,-41.8	Favored (94.1%) <i>m</i> chi angles: 290.8	0.01Å	Favored (96.103%) alpha helix	-	-	-
B 73	LYS	73.07	-	Favored (88.72%) General / -66.3,-38.8	Favored (97.7%) <i>mttt</i> chi angles: 290.4,181,177.9,177.5	0.02Å	Favored (96.765%) alpha helix	-	-	-
B 74	HIS	73.41	-	Favored (92.24%) General / -65.7,-41.5	Favored (88.8%) <i>t70</i> chi angles: 182,73.2	0.01Å	Favored (94.898%) alpha helix	-	-	-
B 75	ASN	71.15	-	Favored (87.67%) General / -60.1,-39.8	Favored (86.5%) <i>m-40</i> chi angles: 285.6,348.6	0.02Å	Favored (91.334%) alpha helix	OUTLIER(S) worst is CG-- OD1: 4.1 σ	-	-
B 76	ASP	70.47	0.80Å OD2 with C 32 ARG HG3	Favored (95.07%) General / -60.2,-44.5	Favored (64%) <i>m-30</i> chi angles: 280.5,341.3	0.01Å	Favored (93.959%) alpha helix	-	-	-
B 77	GLU	73.63	-	Favored (94.72%) General / -62.4,-45.1	Favored (85.2%) <i>tt0</i> chi angles: 178.1,181.8,357.4	0.02Å	Favored (86.686%) alpha helix	-	-	-
B 78	VAL	70.16	-	Favored (87.14%) Ile or Val / -67.2,-41.7	Favored (84.5%) <i>t</i> chi angles: 176.3	0.05Å	Favored (87.58%) alpha helix	-	-	-
B 79	LEU	68.5	0.90Å HD23 with C 31 LYS HD3	Favored (94.28%) General / -63.4,-39.4	Favored (59.5%) <i>tp</i> chi angles: 179.4,57.2	0.02Å	Favored (90.421%) alpha helix	-	-	-
B 80	PHE	70.44	0.72Å HE1 with C 31 LYS HD2	Favored (99.53%) General / -62.2,-42.7	Favored (25%) <i>m-10</i> chi angles: 296.8,341.8	0.01Å	Favored (97.583%) alpha helix	-	-	-
B 81	ARG	70.74	0.47Å NH1 with A 83 GLU OE2	Favored (95.86%) General / -63.4,-44.0	Favored (18.2%) <i>ttm110</i> chi angles: 190,184.4,316,128.7	0.03Å	Favored (95.578%) alpha helix	-	-	-
B 82	LEU	69.42	1.06Å HD21 with A 83 GLU HA	Favored (88.61%) General / -66.2,-38.7	Favored (80.3%) <i>mt</i> chi angles: 289.8,173.5	0.01Å	Favored (92.746%) alpha helix	-	-	-
B 83	GLU	69.18	-	Favored (99.67%) General / -63.0,-41.9	Favored (89.7%) <i>tt0</i> chi angles: 181.9,180.4,7.2	0.01Å	Favored (96.847%) alpha helix	-	-	-
B 84	ASN	70.59	-	Favored (96.42%) General / -64.2,-42.6	Favored (97.3%) <i>m-40</i> chi angles: 291,343	0.01Å	Favored (96.454%) alpha helix	-	-	-
B 85	HIS	69.18	-	Favored (96.35%) General / -64.6,-41.0	Favored (13.9%) <i>m-70</i> chi angles: 287.3,323	0.01Å	Favored (97.545%) alpha helix	-	-	-
B 86	ALA	69.22	-	Favored (93.68%) General / -63.8,-44.3	-	0.02Å	Favored (94.668%) alpha helix	-	-	-
B 87	GLU	71.08	-	Favored (93.59%) General / -65.0,-39.5	Favored (70.6%) <i>tp30</i> chi angles: 182.3,66.2,18.9	0.02Å	Favored (92.438%) alpha helix	-	-	-

B 88	THR	70.36	1.14Å HG21 with C 20 PHE HA	Favored (84.71%) General / -67.6,-40.1	Favored (15.8%) <i>m</i> chi angles: 310	0.09Å	Favored (98.914%) alpha helix	OUTLIER(S) worst is CB--CG2: 4.6 σ	-	-	
B 89	LEU	70.49	0.76Å CD2 with C 20 PHE HE1	Favored (84.84%) General / -66.0,-37.0	Favored (63.1%) <i>mt</i> chi angles: 286.1,169.5	0.01Å	Favored (87.932%) alpha helix	-	-	-	
B 90	ARG	69.3	-	Favored (86.73%) General / -60.5,-47.3	Favored (78.5%) <i>ttp80</i> chi angles: 184,191.2,63.9,84	0.01Å	Favored (83.013%) alpha helix	-	-	-	
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327
B 91	ALA	71.94	-	Favored (95.76%) General / -63.4,-39.9	-	0.03Å	Favored (87.888%) alpha helix	-	-	-	
B 92	ALA	68.9	-	Favored (92.67%) General / -64.2,-38.9	-	0.02Å	Favored (90.016%) alpha helix	-	-	-	
B 93	MET	69.86	-	Favored (96.62%) General / -63.4,-40.2	Favored (70.5%) <i>mmm</i> chi angles: 284.1,288.9,291.4	0.02Å	Favored (95.9%) alpha helix	-	-	-	
B 94	ILE	71.45	-	Favored (94.77%) Ile or Val / -63.8,-46.3	Favored (91.3%) <i>mt</i> chi angles: 293.7,172.7	0.06Å	Favored (89.474%) alpha helix	-	-	-	
B 95	SER	69.86	0.99Å HB3 with C 14 ILE CD1	Favored (92.97%) General / -64.8,-39.0	Favored (70.9%) <i>m</i> chi angles: 296.2	0.01Å	Favored (91.054%) alpha helix	-	-	-	
B 96	LEU	68.61	-	Favored (87.44%) General / -66.9,-39.2	Favored (48.5%) <i>tp</i> chi angles: 183.7,58.9	0.02Å	Favored (91.952%) alpha helix	-	-	-	
B 97	ALA	70.14	-	Favored (98.05%) General / -61.0,-42.2	-	0.02Å	Favored (94.753%) alpha helix	-	-	-	
B 98	LYS	71.67	-	Favored (96.57%) General / -64.4,-41.5	Favored (62%) <i>mttm</i> chi angles: 292.5,180.5,178.6,293.8	0.02Å	Favored (95.921%) alpha helix	-	-	-	
B 99	LYS	68.18	0.64Å NZ with A 109 TYR CE2	Favored (99.11%) General / -63.2,-41.3	Favored (82.5%) <i>tttt</i> chi angles: 178.3,182.1,177.7,184.6	0.02Å	Favored (96.435%) alpha helix	-	-	-	
B 100	ILE	68.58	-	Favored (98.15%) Ile or Val / -63.6,-43.4	Favored (82.1%) <i>mt</i> chi angles: 290.3,169.7	0.05Å	Favored (98.75%) alpha helix	-	-	-	
B 101	ASP	69.64	-	Favored (97.91%) General / -61.7,-43.9	Favored (33.3%) <i>m-30</i> chi angles: 274.1,356.7	0.00Å	Favored (94.317%) alpha helix	-	-	-	
B 102	VAL	69.4	0.91Å HG21 with C 12 LEU CD1	Favored (94.23%) Ile or Val / -65.5,-44.2	Favored (58.2%) <i>t</i> chi angles: 170.5	0.05Å	Favored (89.899%) alpha helix	-	-	-	

B 103	GLN 69.72			1.33Å O with B 104 THR N	Favored (61.76%) General / -72.2,-19.3	Favored (16.5%) <i>mt0</i> chi angles: 291.7,184.3,167.7	0.02Å	Favored (56.51%) alpha helix	OUTLIER(S) worst is C--O: 24.5 σ	OUTLIER(S) worst is O-C-N: 12.7 σ	-
B 104		THR	67.87	1.33Å N with B 103 GLN O	Favored (38.8%) General / -103.0,8.6	Favored (59.7%) <i>p</i> chi angles: 57.3	0.05Å	Favored (50.396%)	-	-	-
B 105	GLY 68.21			-	Favored (63.74%) Glycine / 82.8,19.3	-	-	Favored (84.721%)	-	-	-
B 106		ARG	65	-	Favored (21.53%) General / -112.9,157.4	Favored (72.4%) <i>mtp180</i> chi angles: 299.4,187.1,59.1,188.9	0.01Å	Favored (23.72%)	-	-	-
B 107	ARG 58.98			-	Favored (87.56%) Pre-Pro / -74.4,151.5	Favored (44.2%) <i>ptt-90</i> chi angles: 72.4,185.3,191.4,275.7	0.00Å	Favored (39.201%)	-	-	-
B 108		PRO	55.58	-	Favored (57.72%) Trans-Pro / -65.6,157.6	Favored (34.5%) <i>Cg_exo</i> chi angles: 339.4,33.2,327.9	0.04Å	Favored (67.276%)	-	-	-
B 109	TYR 56.97			-	Favored (46.95%) General / -69.8,150.3	Favored (37.3%) <i>m-80</i> chi angles: 288.8,301.4	0.01Å	-	-	-	-
B 110		GLU	52.7	-	-	Favored (6.6%) <i>tt0</i> chi angles: 205.4,172.6,35.1	0.02Å	-	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327
C 1	MET 34.64			0.72Å SD with C 4 THR O	-	Favored (46.2%) <i>ttm</i> chi angles: 186.4,175.6,277	0.01Å	-	-	-	-
C 2		ASP	42.31	-	Favored (19.27%) General / -90.3,12.1	Favored (46.9%) <i>p0</i> chi angles: 58.9,4	0.02Å	-	-	-	-
C 3	GLY 43.38			-	Favored (30.57%) Glycine / 109.7,-1.8	-	-	Favored (45.589%)	-	-	-
C 4		THR	44.02	0.72Å O with C 1 MET SD	Favored (21.81%) General / -105.6,153.4	Favored (65.9%) <i>p</i> chi angles: 63	0.02Å	Favored (33.128%)	-	-	-
C 5	LEU 44.56			0.99Å HD12 with A 104 THR O	Favored (36.61%) General / -80.8,-27.6	Favored (52.4%) <i>mt</i> chi angles: 287.7,176.1	0.00Å	Favored (38.086%)	-	-	-
C 6		PHE	43.32	0.66Å CG with B 103 GLN HA	Favored (27.9%) Pre-Pro / -126.6,135.9	Favored (45.2%) <i>m-80</i> chi angles: 308.8,283.7	0.01Å	Favored (23.865%)	-	-	-
C 7	PRO 47.35			-	Favored (64.5%) Trans-Pro / -62.9,137.7	Favored (36.2%) <i>Cg_exo</i> chi angles: 338.9,33.2,328.5	0.03Å	Favored (28.32%)	-	-	-
C 8		GLY	47.85	-	Favored (87.17%) Glycine / 86.6,-1.3	-	-	Favored (81.387%)	-	-	-
C 9	ASP 51.58			0.59Å OD1 with B	Favored (93.13%)	Favored (18.3%) <i>m-30</i>	0.00Å	Favored (39.936%)	-	-	-

				103 GLN HG2	General / -59.8,-42.4	chi angles: 271.3,357.8	alpha helix				
C 10	ASP	53.08	-	-	Favored (85.3%) General / -65.1,-36.8	Favored (93.9%) <i>m</i> - 30 chi angles: 289.3,340.5	0.01Å	Favored (94.108%) alpha helix	-	-	-
C 11	ASP	53.52	-	-	Favored (67.67%) General / -69.7,-30.1	Favored (95.1%) <i>m</i> - 30 chi angles: 292,341.6	0.01Å	Favored (77.801%) alpha helix	-	-	-
C 12	LEU	53.69	0.91Å CD1 with B 102 VAL HG21	-	Favored (46.29%) General / -98.5,7.3	Favored (86.2%) <i>mt</i> chi angles: 299.2,179	0.01Å	Favored (38.47%)	-	-	-
C 13	ALA	54.86	-	-	Favored (50.2%) General / -64.8,132.7	-	0.02Å	Favored (24.144%)	-	-	-
C 14	ILE	51.35	0.99Å CD1 with B 95 SER HB3	-	Favored (17.38%) Pre-Pro / -130.5,116.4	Favored (59.7%) <i>mt</i> chi angles: 304.2,175.1	0.01Å	Favored (51.538%) beta sheet	-	-	-
C 15	PRO	55.42	-	-	Favored (86.97%) Trans-Pro / -62.1,141.0	Favored (54.9%) <i>Cg_exo</i> chi angles: 337.4,33.9,328.7	0.03Å	Favored (63.514%)	-	-	-
C 16	ALA	50.63	-	-	Favored (71.63%) General / -60.2,-32.9	-	0.02Å	Favored (58.673%)	-	-	-
C 17	THR	53.89	-	-	Favored (67.7%) General / -64.6,-24.8	Favored (42.4%) <i>p</i> chi angles: 55	0.03Å	Favored (70.381%) three-ten	-	-	-
C 18	GLU	55.72	-	-	Favored (59.66%) General / -81.9,-8.2	Favored (90.1%) <i>mt-10</i> chi angles: 294.9,182.7,8.8	0.01Å	Favored (58.176%) three-ten	-	-	-
C 19	PHE	50.87	0.93Å O with B 88 THR HG22	-	Favored (65.16%) General / -67.9,-21.5	Favored (7.9%) <i>m</i> - 10 chi angles: 279.2,15	0.07Å	Favored (43.511%)	-	OUTLIER(S) worst is CA-C- O: 5.7 σ	-
C 20	PHE	48.38	1.24Å CE1 with A 90 ARG HG3	-	Favored (2.16%) General / -136.2,48.5	Favored (51.4%) <i>m</i> - 80 chi angles: 307.6,281.1	0.03Å	Favored (16.312%)	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non- Trans: 0 of 327
C 21	SER	46.91	-	-	Favored (28.83%) General / -79.8,154.0	Favored (45.2%) <i>p</i> chi angles: 56.1	0.02Å	Favored (18.139%)	-	-	-
C 22	THR	43.71	-	-	Favored (45.49%) General / -72.4,134.6	Favored (99.8%) <i>m</i> chi angles: 300.5	0.05Å	Favored (35.58%)	-	-	-
C 23	LYS	43.95	-	-	Favored (11.52%) General / -102.9,164.6	Favored (28%) <i>mmtm</i> chi angles: 298.4,291.3,180.6,285.6	0.01Å	Favored (21.54%)	-	-	-
C 24	ALA	42.37	-	-	Favored (55.74%) General / -68.3,141.6	-	0.02Å	Favored (34.003%)	-	-	-

C 25	ALA	45.53	-		Favored (15.96%) General / -100.3,158.3	-	0.01Å	Favored (39.396%)	-	-	-
C 26	LYS	49.83	0.55Å HB2 with C 31 LYS HE3		Favored (15.33%) General / -73.7,170.4	Favored (84.6%) <i>mttt</i> chi angles: 280.3,182.6,177.4,176.7	0.01Å	Favored (12.206%)	-	-	-
C 27	ASN	54.38	-		Favored (66.15%) Pre-Pro / -60.8,122.1	Favored (58%) <i>t0</i> chi angles: 187,352.2	0.04Å	Favored (15.786%)	-	-	-
C 28	PRO	57.92	0.40Å HA with B 80 PHE HZ		Favored (74.77%) Trans-Pro / -63.0,-33.4	Favored (39.5%) <i>Cg_exo</i> chi angles: 338.4,33.2,329.1	0.02Å	Favored (73.909%)	-	-	-
C 29	GLU	62.6	-		Favored (91.87%) General / -63.4,-38.6	Favored (68.8%) <i>tp30</i> chi angles: 183,65.9,21	0.01Å	Favored (74.065%) alpha helix	-	-	-
C 30	THR	65.23	-		Favored (99.19%) General / -61.3,-43.2	Favored (97.7%) <i>m</i> chi angles: 300	0.04Å	Favored (90.621%) alpha helix	-	-	-
C 31	LYS	61.34	0.90Å HD3 with B 79 LEU HD23		Favored (75.89%) General / -70.0,-38.5	Favored (98.6%) <i>mttt</i> chi angles: 295,177.4,179.6,174	0.02Å	Favored (91.992%) alpha helix	-	-	-
C 32	ARG	64.63	0.80Å HG3 with B 76 ASP OD2		Favored (97.77%) General / -63.9,-41.5	Favored (96.6%) <i>mtm-85</i> chi angles: 284.7,191.4,289.3,271	0.00Å	Favored (98.154%) alpha helix	-	-	-
C 33	GLU	68.59	-		Favored (87.59%) General / -65.1,-37.5	Favored (99.9%) <i>mt-10</i> chi angles: 291.9,178.3,353.6	0.01Å	Favored (89.009%) alpha helix	-	-	-
C 34	ALA	64.92	-		Favored (90.78%) General / -63.4,-38.3	-	0.02Å	Favored (85.146%) alpha helix	-	-	-
C 35	ILE	63.52	-		Favored (78.43%) Ile or Val / -69.8,-45.0	Favored (95.5%) <i>mt</i> chi angles: 293.5,170	0.06Å	Favored (81.066%) alpha helix	-	-	-
C 36	VAL	63.39	0.72Å HG21 with C 44 GLU HB2		Favored (51.23%) Ile or Val / -65.2,-30.7	Favored (74.1%) <i>t</i> chi angles: 172.6	0.06Å	Favored (78.128%) alpha helix	-	-	-
C 37	LYS	63.94	0.76Å HE3 with C 43 ASN HB3		Favored (27.48%) General / -89.9,9.5	Favored (98%) <i>mttt</i> chi angles: 291.6,182.4,175.6,179.7	0.01Å	Favored (31.695%)	-	-	-
C 38	ALA	55.78	-		Favored (88.29%) General / -60.8,-39.3	-	0.02Å	Favored (13.63%)	-	-	-
C 39	TYR	55.77	0.68Å HE2 with A 71 CYS HB2		Favored (26.94%) General / -130.8,122.3	Favored (86.9%) <i>m-80</i> chi angles: 300.9,280.7	0.03Å	CaBLAM Disfavored (2.007%)	-	-	-
C 40	GLY	58.36	-		Favored (37.05%) Glycine / 95.5,174.5	-	-	Favored (34.175%)	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327

C 41	ASP	58.94	-		Favored (25.37%) General / -91.4,-17.7	Favored (70.1%) <i>m-30</i> chi angles: 298.9,332.3	0.02Å	CaBLAM Disfavored (3.941%)	-	-	-
C 42	ASP	60.81	1.01Å OD2 with C 50 ARG NH1		Favored (22.42%) General / -88.2,151.8	Favored (78.2%) <i>m-30</i> chi angles: 295.5,328.1	0.01Å	Favored (18.837%)	-	-	-
C 43	ASN	64.46	0.76Å HB3 with C 37 LYS HE3		Favored (3.44%) General / -102.3,-177.6	Favored (44.7%) <i>p0</i> chi angles: 60.3,12.9	0.04Å	Favored (27.506%)	-	-	-
C 44	GLU	66.86	0.72Å HB2 with C 36 VAL HG21		Favored (94.41%) General / -62.6,-45.1	Favored (64.8%) <i>tt0</i> chi angles: 179.2,183.6,336	0.03Å	Favored (44.433%)	-	-	-
C 45	GLU	71.5	-		Favored (90.53%) General / -65.9,-39.3	Favored (71.7%) <i>tp30</i> chi angles: 179.7,65.8,15.6	0.01Å	Favored (91.983%) alpha helix	-	-	-
C 46	THR	74.33	-		Favored (92.76%) General / -65.3,-39.7	Favored (88.5%) <i>m</i> chi angles: 298.4	0.04Å	Favored (95.877%) alpha helix	-	-	-
C 47	LEU	75.31	-		Favored (96.73%) General / -63.8,-40.5	Favored (75.7%) <i>mt</i> chi angles: 288,171.8	0.01Å	Favored (96.754%) alpha helix	-	-	-
C 48	LYS	74.78	0.59Å HG3 with B 65 PHE CZ		Favored (98.23%) General / -61.8,-43.8	Favored (97.3%) <i>mttt</i> chi angles: 290.7,182.7,178.6,179.7	0.01Å	Favored (96.724%) alpha helix	-	-	-
C 49	GLN	80.1	-		Favored (93.4%) General / -65.3,-40.2	Favored (95.1%) <i>mt0</i> chi angles: 294.1,172.1,343.9	0.01Å	Favored (96.724%) alpha helix	-	-	-
C 50	ARG	79.54	1.01Å NH1 with C 42 ASP OD2		Favored (93.02%) General / -65.4,-42.0	Favored (88.5%) <i>mtp180</i> chi angles: 290.9,175.3,65,188.7	0.02Å	Favored (98.805%) alpha helix	-	-	-
C 51	LEU	79.37	0.70Å HD21 with B 61 ILE HG21		Favored (97.36%) General / -63.9,-42.7	Favored (60.8%) <i>mt</i> chi angles: 285.7,168.6	0.01Å	Favored (98.554%) alpha helix	-	-	-
C 52	THR	81.81	-		Favored (99.75%) General / -62.9,-42.3	Favored (88.5%) <i>m</i> chi angles: 298.4	0.05Å	Favored (99.061%) alpha helix	-	-	-
C 53	ASN	81.58	-		Favored (96.04%) General / -64.8,-40.9	Favored (99%) <i>m-40</i> chi angles: 289.1,340.7	0.02Å	Favored (98.299%) alpha helix	-	-	-
C 54	LEU	80.07	-		Favored (87.41%) General / -66.5,-38.5	Favored (5.9%) <i>mp</i> chi angles: 265.6,61.5	0.02Å	Favored (97.231%) alpha helix	-	-	-
C 55	GLU	81.79	0.81Å HG2 with B 58 ILE HG21		Favored (98.51%) General / -61.5,-42.2	Favored (97.7%) <i>mt-10</i> chi angles: 291.1,179.4,3.5	0.01Å	Favored (97.333%) alpha helix	-	-	-
C 56	LYS	83.66	-		Favored (95.19%) General / -65.1,-41.3	Favored (98.5%) <i>mttt</i> chi angles: 292.4,178.6,180.8,177.7	0.01Å	Favored (98.509%) alpha helix	-	-	-
C 57	LYS	78.99	-		Favored (94.95%) General / -64.6,-43.1	Favored (63%) <i>mttm</i> chi angles: 292.3,180.2,184.3,293.5	0.01Å	Favored (98.648%) alpha helix	-	-	-

C 58	ILE	78.19	0.76Å HD12 with B 58 ILE HG13	Favored (83.44%) Ile or Val / -66.2,-39.1	Favored (68.9%) <i>mt</i> chi angles: 291.4,175.3	0.06Å	Favored (96.041%) alpha helix	-	-	-	
C 59	THR	81.29	-	Favored (99.41%) General / -62.6,-43.2	Favored (88.5%) <i>m</i> chi angles: 298.6	0.05Å	Favored (97.901%) alpha helix	-	-	-	
C 60	ASN	78.81	-	Favored (94.74%) General / -65.1,-40.4	Favored (98.6%) <i>m-40</i> chi angles: 290.6,338.6	0.01Å	Favored (96.615%) alpha helix	-	-	-	
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327
C 61	ILE	74.89	-	Favored (88.61%) Ile or Val / -66.0,-40.9	Favored (95.9%) <i>mt</i> chi angles: 293,170	0.06Å	Favored (98.424%) alpha helix	-	-	-	
C 62	THR	77.2	-	Favored (98.92%) General / -61.8,-42.4	Favored (98.3%) <i>m</i> chi angles: 300.6	0.04Å	Favored (98.169%) alpha helix	-	-	-	
C 63	THR	77.92	-	Favored (96.56%) General / -62.5,-44.4	Favored (92.1%) <i>m</i> chi angles: 299	0.05Å	Favored (98.834%) alpha helix	-	-	-	
C 64	LYS	73.24	-	Favored (99.31%) General / -63.1,-41.5	Favored (93.8%) <i>mttt</i> chi angles: 288,186.1,176.5,182.3	0.01Å	Favored (97.169%) alpha helix	-	-	-	
C 65	PHE	73.28	-	Favored (97.23%) General / -64.1,-41.4	Favored (78.3%) <i>t80</i> chi angles: 183.2,84.6	0.01Å	Favored (97.626%) alpha helix	-	-	-	
C 66	GLU	75.97	-	Favored (90.49%) General / -64.1,-38.3	Favored (99.9%) <i>mt-10</i> chi angles: 292.1,175.8,350	0.02Å	Favored (95.26%) alpha helix	-	-	-	
C 67	GLN	74.32	-	Favored (83.73%) General / -67.9,-39.2	Favored (99.4%) <i>mt0</i> chi angles: 291.9,176.7,326.2	0.01Å	Favored (99.496%) alpha helix	-	-	-	
C 68	ILE	69.18	-	Favored (89.7%) Ile or Val / -65.8,-41.2	Favored (95.5%) <i>mt</i> chi angles: 292.4,170.3	0.05Å	Favored (97.513%) alpha helix	-	-	-	
C 69	GLU	70.78	0.42Å OE2 with B 48 LYS NZ	Favored (99.66%) General / -62.8,-42.3	Favored (81.3%) <i>tt0</i> chi angles: 182.5,183.4,8.9	0.01Å	Favored (98.085%) alpha helix	-	-	-	
C 70	LYS	73.74	-	Favored (97.61%) General / -61.0,-42.0	Favored (83.7%) <i>tttt</i> chi angles: 178.2,180.1,177,181.5	0.03Å	Favored (98.404%) alpha helix	-	-	-	
C 71	CYS	68.72	-	Favored (98.1%) General / -62.5,-43.7	Favored (38.6%) <i>t</i> chi angles: 175.4	0.03Å	Favored (99.044%) alpha helix	-	-	-	
C 72	CYS	68.14	0.52Å SG with A 35 ILE CG2	Favored (98.27%) General / -63.6,-41.8	Favored (92.9%) <i>m</i> chi angles: 290.9	0.01Å	Favored (95.959%) alpha helix	-	-	-	

C 73	LYS	69.71	-		Favored (85.09%) General / -67.5,-38.8	Favored (97.7%) <i>mttt</i> chi angles: 290.2,181.6,177.5,177.1	0.02Å	Favored (95.539%) alpha helix	-	-	-
C 74	HIS	70.42	-		Favored (88.29%) General / -66.6,-41.4	Favored (88%) <i>t70</i> chi angles: 182.6,73.1	0.01Å	Favored (96.78%) alpha helix	-	-	-
C 75	ASN	67.26	0.48Å OD1 with A 35 ILE HG12		Favored (89.1%) General / -62.7,-38.1	Favored (6.4%) <i>m110</i> chi angles: 285.9,170.6	0.05Å	Favored (92.075%) alpha helix	-	-	-
C 76	ASP	66.24	-		Favored (98.21%) General / -61.3,-43.8	Favored (58.7%) <i>m-30</i> chi angles: 279.6,341.5	0.01Å	Favored (93.843%) alpha helix	-	-	-
C 77	GLU	69.86	-		Favored (94.9%) General / -63.0,-44.9	Favored (83.7%) <i>tt0</i> chi angles: 178,182.6,357.8	0.03Å	Favored (91.331%) alpha helix	-	-	-
C 78	VAL	66.37	-		Favored (92.78%) Ile or Val / -65.6,-42.4	Favored (96.7%) <i>t</i> chi angles: 175	0.05Å	Favored (91.749%) alpha helix	-	-	-
C 79	LEU	64.31	-		Favored (97.34%) General / -63.5,-40.5	Favored (65.4%) <i>tp</i> chi angles: 178.3,58.1	0.02Å	Favored (93.419%) alpha helix	-	-	-
C 80	PHE	65.82	0.75Å HE1 with B 31 LYS NZ		Favored (98.3%) General / -63.6,-42.8	Favored (25.9%) <i>m-10</i> chi angles: 297,340.5	0.01Å	Favored (98.913%) alpha helix	-	-	-
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non- Trans: 0 of 327
C 81	ARG	67.38	-		Favored (93.96%) General / -63.1,-45.1	Favored (16.3%) <i>ttm110</i> chi angles: 189.3,186.1,317.5,128.8	0.03Å	Favored (95.29%) alpha helix	-	-	-
C 82	LEU	63.55	-		Favored (86.48%) General / -66.0,-37.6	Favored (87.7%) <i>mt</i> chi angles: 291.3,174	0.01Å	Favored (91.619%) alpha helix	-	-	-
C 83	GLU	64.32	-		Favored (96.27%) General / -64.5,-41.9	Favored (89.1%) <i>tt0</i> chi angles: 181.6,180.5,7.8	0.01Å	Favored (96.19%) alpha helix	-	-	-
C 84	ASN	65.9	-		Favored (93.05%) General / -65.2,-43.1	Favored (98.6%) <i>m-40</i> chi angles: 289.9,342	0.01Å	Favored (94.922%) alpha helix	-	-	-
C 85	HIS	64.8	-		Favored (94.05%) General / -65.2,-40.4	Favored (14.6%) <i>m-70</i> chi angles: 287.2,322.3	0.00Å	Favored (97.021%) alpha helix	-	-	-
C 86	ALA	64.02	-		Favored (91.81%) General / -65.0,-43.7	-	0.02Å	Favored (95.997%) alpha helix	-	-	-
C 87	GLU	67.1	-		Favored (91.39%) General / -65.8,-39.8	Favored (71.4%) <i>tp30</i> chi angles: 181.7,66,18.8	0.02Å	Favored (92.048%) alpha helix	-	-	-

C 88	THR	65.76	0.59Å HG22 with B 19 PHE O	Favored (79.27%) General / -68.9,-40.2	Favored (8.1%) <i>m</i> chi angles: 313	0.04Å	Favored (97.897%) alpha helix	OUTLIER(S) worst is CB-- CG2: 13.6 σ	OUTLIER(S) worst is OG1- CB-CG2: 5.5 σ	-	
C 89	LEU	65.6	-	Favored (80.12%) General / -66.1,-35.3	Favored (67.4%) <i>mt</i> chi angles: 286.7,170.2	0.01Å	Favored (85.792%) alpha helix	-	-	-	
C 90	ARG	65.11	-	Favored (87.37%) General / -62.1,-46.8	Favored (79%) <i>ttp80</i> chi angles: 184.1,191.9,64.8,83.2	0.02Å	Favored (82.078%) alpha helix	-	-	-	
C 91	ALA	68.78	-	Favored (95.36%) General / -64.3,-40.1	-	0.03Å	Favored (88.232%) alpha helix	-	-	-	
C 92	ALA	65.4	-	Favored (93.46%) General / -64.6,-39.3	-	0.03Å	Favored (92.354%) alpha helix	-	-	-	
C 93	MET	66.61	0.58Å HB3 with A 20 PHE HZ	Favored (93.69%) General / -64.3,-39.3	Favored (70.4%) <i>mmm</i> chi angles: 285.1,288.2,292	0.01Å	Favored (96.659%) alpha helix	-	-	-	
C 94	ILE	68.46	-	Favored (94.14%) Ile or Val / -63.9,-46.5	Favored (87.7%) <i>mt</i> chi angles: 293.2,173.3	0.06Å	Favored (88.75%) alpha helix	-	-	-	
C 95	SER	67.68	1.04Å HB3 with B 14 ILE HD11	Favored (93.4%) General / -64.9,-39.4	Favored (68.9%) <i>m</i> chi angles: 296.6	0.02Å	Favored (90.326%) alpha helix	-	-	-	
C 96	LEU	67.1	-	Favored (87.58%) General / -66.8,-39.0	Favored (48.9%) <i>tp</i> chi angles: 183.7,59.1	0.02Å	Favored (91.241%) alpha helix	-	-	-	
C 97	ALA	67.7	-	Favored (98.33%) General / -61.8,-41.9	-	0.02Å	Favored (93.405%) alpha helix	-	-	-	
C 98	LYS	71.55	-	Favored (95.64%) General / -65.0,-41.1	Favored (62.5%) <i>mttm</i> chi angles: 292.2,180.6,179.4,294.9	0.02Å	Favored (96.611%) alpha helix	-	-	-	
C 99	LYS	67.34	0.72Å HA with B 12 LEU HD13	Favored (97.51%) General / -64.0,-41.0	Favored (83.4%) <i>tttt</i> chi angles: 178.8,180.8,179,181.6	0.01Å	Favored (95.608%) alpha helix	-	-	-	
C 100	ILE	68.83	-	Favored (98.1%) Ile or Val / -63.4,-43.0	Favored (78.1%) <i>mt</i> chi angles: 289.9,170.4	0.06Å	Favored (98.034%) alpha helix	-	-	-	
#	Alt	Res	High B	Clash > 0.4Å	Ramachandran	Rotamer	Cβ deviation	CaBLAM	Bond lengths	Bond angles	Cis Peptides
			Avg: 63.29	Clashscore: 30.78	Outliers: 0 of 324	Poor rotamers: 0 of 288	Outliers: 0 of 318	Outliers: 0 of 318	Outliers: 5 of 330	Outliers: 3 of 330	Non-Trans: 0 of 327
C 101	ASP	68.79	-	Favored (99.39%) General / -61.6,-43.1	Favored (27.7%) <i>m-30</i> chi angles: 273.3,357.2	0.01Å	Favored (96.444%) alpha helix	-	-	-	
C 102	VAL	68.36	0.58Å HG12 with B 7 PRO O	Favored (91.05%) Ile or Val / -66.4,-44.0	Favored (62.9%) <i>t</i> chi angles: 171.2	0.06Å	Favored (91.866%) alpha helix	-	-	-	

C 103	GLN 70.11	0.67Å HG2 with B 9 ASP OD1	Favored (63.39%) General / -70.0,-24.9	Favored (16.3%) <i>mt0</i> chi angles: 289.7,182.5,169.4	0.03Å	Favored (71.188%) alpha helix	-	-	-
C 104	THR 67.21	-	Favored (44.02%) General / -98.5,9.1	Favored (68.3%) <i>p</i> chi angles: 58.9	0.03Å	Favored (51.62%)	-	-	-
C 105	GLY 67.34	-	Favored (61.4%) Glycine / 84.5,18.9	-	-	Favored (84.156%)	-	-	-
C 106	ARG 63.06	-	Favored (24.55%) General / -112.4,155.2	Favored (73.6%) <i>mtp180</i> chi angles: 298.1,188.1,60.7,188.7	0.01Å	Favored (24.67%)	-	-	-
C 107	ARG 58.08	-	Favored (89.25%) Pre-Pro / -73.5,150.7	Favored (43.6%) <i>ptt-90</i> chi angles: 71.1,185.2,192,273.9	0.01Å	Favored (39.996%)	-	-	-
C 108	PRO 56.62	-	Favored (50.82%) Trans-Pro / -64.7,158.3	Favored (35.4%) <i>Cg_exo</i> chi angles: 339.2,33.3,327.9	0.04Å	Favored (63.756%)	-	-	-
C 109	TYR 54.21	0.46Å CD1 with A 14 ILE O	Favored (49.13%) General / -68.8,149.3	Favored (36.4%) <i>m-80</i> chi angles: 289.1,302.3	0.01Å	-	-	-	-
C 110	GLU 45.13	0.70Å O with A 15 PRO HB3	-	Allowed (0.3%) <i>mt-10</i> chi angles: 240.9,176.7,13.9	0.02Å	-	-	-	-

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